

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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JAVA ASSIGNMENT

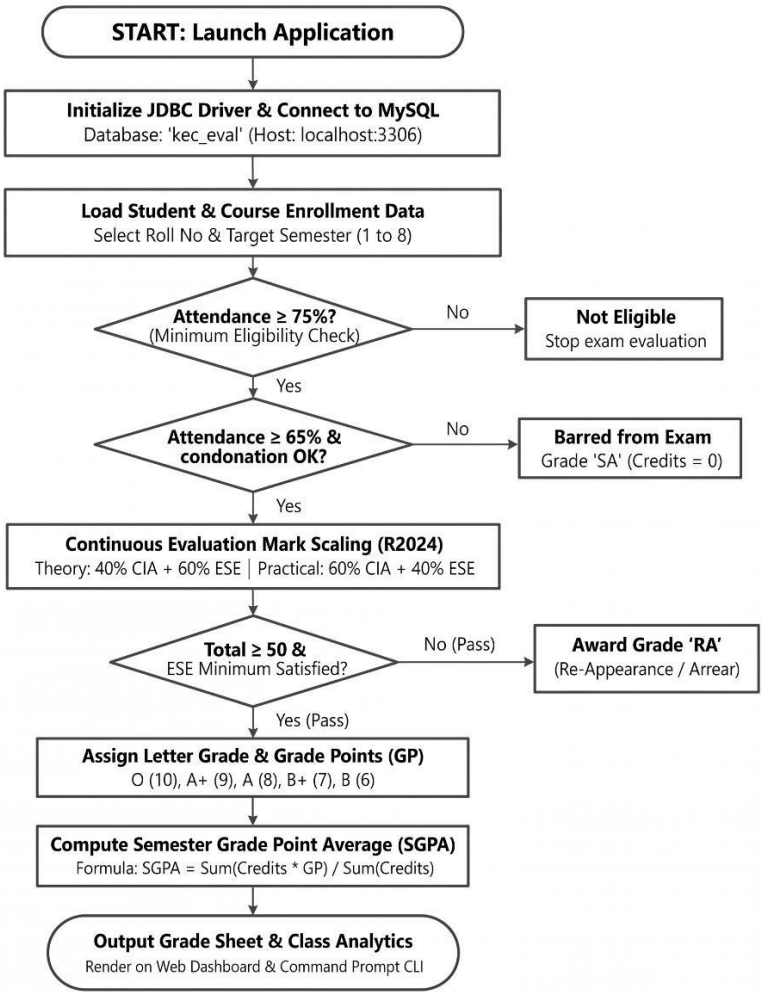
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
mysql> describe students;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| roll_no          | varchar(10)  | NO   | PRI | NULL    |       |
| student_name     | varchar(100) | NO   |     | NULL    |       |
| department       | varchar(10)  | NO   |     | NULL    |       |
| current_semester | int          | NO   |     | NULL    |       |
| email            | varchar(100) | YES  |     | NULL    |       |
| phone            | varchar(15)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. COURSES TABLE:

```
mysql> describe courses;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| course_code | varchar(10)  | NO   | PRI | NULL    |       |
| title       | varchar(100) | NO   |     | NULL    |       |
| credits     | decimal(3,1) | NO   |     | NULL    |       |
| course_type | varchar(10)  | NO   |     | NULL    |       |
| department  | varchar(10)  | NO   |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+------------------+-------------+------+-----+---------+----------------+
| Field            | Type        | Null | Key | Default | Extra          |
+------------------+-------------+------+-----+---------+----------------+
| enrollment_id    | int         | NO   | PRI | NULL    | auto_increment |
| roll_no          | varchar(10) | NO   | MUL | NULL    |                |
| course_code      | varchar(10) | NO   | MUL | NULL    |                |
| semester         | int         | NO   |     | NULL    |                |
| classes_held     | int         | NO   |     | NULL    |                |
| classes_attended | int         | NO   |     | NULL    |                |
+------------------+-------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| mark_id         | int         | NO   | PRI | NULL    | auto_increment |
| enrollment_id   | int         | NO   | MUL | NULL    |                |
| cia1            | decimal(5,2) | YES  |     | NULL    |                |
| cia2            | decimal(5,2) | YES  |     | NULL    |                |
| assignment_mark | decimal(5,2) | YES  |     | NULL    |                |
| ese_mark        | decimal(5,2) | YES  |     | NULL    |                |
| ese_absent      | tinyint(1)  | YES  |     | 0       |                |
+-----------------+-------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.sql.Connection;

import java.sql.SQLException;

import java.util.Scanner;


public class MainApp {

    private static final Scanner sc = new Scanner(System.in);

    private static final StudentDAO students = new StudentDAO();

    private static final AcademicDAO academic = new AcademicDAO();


    public static void main(String[] args) {

        try (Connection c = DBConnection.getConnection()) {

            System.out.println("Database connection successful.");

        } catch (SQLException e) {

            System.out.println("Database connection failed: " + e.getMessage());

            System.out.println("Check MySQL is running and update DBConnection.java password.");

            return;

        }


        while (true) {

            System.out.println("\n=== KEC STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM

===");
```

```
            System.out.println("1. Add Student");
            System.out.println("2. Add Course");
            System.out.println("3. Enroll Student in Course");
            System.out.println("4. Update Attendance");
            System.out.println("5. Enter/Update Incremental Marks");
            System.out.println("6. Generate Semester Mark Sheet");
            System.out.println("7. Attendance Defaulter Report");
            System.out.println("0. Exit");
            int choice = readInt("Choose: ");
            try {
                switch (choice) {
                    case 1:
                        students.addStudent(read("Roll number: "), read("Student name: "),
                            read("Department (CSE/ECE/IT/MECH etc.): "), readInt("Current semester: "));
                        break;
                    case 2:
                        academic.addCourse(read("Course code: "), read("Course title: "),
                            readDouble("Credits: "), read("Course type (THEORY/PRACTICAL/EMBEDDED): "));
                        break;
                    case 3:
                        academic.enroll(read("Roll number: "), read("Course code: "), readInt("Semester: "));
                        break;
                    case 4:
                        String ar = read("Roll number: "), ac = read("Course code: ");
                        int asem = readInt("Semester: "), held = readInt("Classes held: "), attended = readInt("Classes
attended: ");
                        boolean cond = read("Medical condonation approved? (y/n): ").equalsIgnoreCase("y");
                        academic.updateAttendance(ar, ac, asem, held, attended, cond);
                        break;
                    case 5:
                        String roll = read("Roll number: "), code = read("Course code: ");
                        int sem = readInt("Semester: ");
                        System.out.println("Enter a mark from 0-100, or press Enter to leave it unchanged.");
```

```
                Double cia1 = readOptionalMark("CIA Test 1: ");
                Double cia2 = readOptionalMark("CIA Test 2: ");
                Double assignment = readOptionalMark("Assignment/Quiz/Lab evaluation: ");
                String absentAnswer = read("Was the student absent for ESE? (y/n): ");
                boolean absent = absentAnswer.equalsIgnoreCase("y");
                Double ese = absent ? null : readOptionalMark("ESE mark: ");
                academic.updateMarks(roll, code, sem, cia1, cia2, assignment, ese, absent);
                break;
            case 6:
                academic.printMarkSheet(read("Roll number: "), readInt("Semester: "));
                break;
            case 7:
                academic.attendanceDefaulters();
                break;
            case 0:
                System.out.println("Exiting."); return;
            default:
                System.out.println("Invalid option.");
          }
       } catch (Exception e) {
          System.out.println("Operation failed: " + e.getMessage());
       }
    }
}

private static String read(String prompt) {
    System.out.print(prompt); return sc.nextLine().trim();
}
private static int readInt(String prompt) {
    while (true) try { return Integer.parseInt(read(prompt)); }
    catch (NumberFormatException e) { System.out.println("Enter a valid integer."); }
}
```

```
    private static double readDouble(String prompt) {
        while (true) try { return Double.parseDouble(read(prompt)); }
        catch (NumberFormatException e) { System.out.println("Enter a valid number."); }
    }
    private static Double readOptionalMark(String prompt) {
        while (true) {
            String s = read(prompt);
            if (s.isEmpty()) return null;
            try {
                double d = Double.parseDouble(s);
                if (d >= 0 && d <= 100) return d;
            } catch (NumberFormatException ignored) {}
            System.out.println("Enter a number from 0 to 100, or press Enter.");
        }
    }
```

DBCONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/kec_academic";
    private static final String USER = "root";
    private static final String PASSWORD = "Subadharsh2008"; // Change this

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
=== KEC STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM ===
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Update Attendance
5. Enter/Update Incremental Marks
6. Generate Semester Mark Sheet
7. Attendance Defaulter Report
0. Exit
Choose:
```

STUDENT & COURSE MANAGEMENT:

1.ADDING STUDENT:

```
Choose: 1
Roll number: 25ITR143
Student name: Subadharsini
Department (CSE/ECE/IT/MECH etc.): IT
Current semester: 3
Student added successfully.
```

2.ADDING COURSE:

```
Choose: 2
Course code: 24ITT32
Course title: Programming in Java
Credits: 3
Course type (THEORY/PRACTICAL/EMBEDDED): Embedded
Course added successfully.
```

3.STUDENT ENROLLMENT IN COURSE:

```
Choose: 3
Roll number: 25ITR143
Course code: 24ITT32
Semester: 3
Student enrolled successfully.
```

4.ATTENDANCE UPDATING FOR STUDENT:

```
Choose: 4
Roll number: 25ITR143
Course code: 24ITT32
Semester: 3
Classes held: 50
Classes attended: 48
Medical condonation approved? (y/n): Y
Attendance updated.
```

5.MARKS UPDATING FOR STUDENT:

```
Choose: 5
Roll number: 25ITR143
Course code: 24ITT32
Semester: 3
Enter a mark from 0-100, or press Enter to leave it unchanged.
CIA Test 1: 88
CIA Test 2: 87
Assignment/Quiz/Lab evaluation: 96
Was the student absent for ESE? (y/n): N
ESE mark: 95
Marks saved/updated. Leave a field blank to keep its previous value.
```

6.MARKSHEET GENERATION:

```
Choose: 6
Roll number: 25ITR143
Semester: 3

=== KONGU ENGINEERING COLLEGE - SEMESTER MARK SHEET ===
Roll No: 25ITR143

Student: Subadharsini | Department: IT
Course: 24ITT32 - Programming in Java (EMBEDDED, 3.0 credits)
Attendance: 96.0% | CIA1: 88.00 | CIA2: 87.00 | Assignment: 96.00 | ESE: 95.00
Weighted Total: 92.67 | Grade: O | Result: PASS

SGPA: 10.00 | Equivalent Percentage (SGPA x 10): 100.00%
Note: This is a project implementation of the rules supplied in the project brief; verify institutional regulations before official use.
```

7.ATTENDANCE REPORT:

```
=== KEC STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM ===
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Update Attendance
5. Enter/Update Incremental Marks
6. Generate Semester Mark Sheet
7. Attendance Defaulter Report
0. Exit
Choose: 7

=== ATTENDANCE DEFAULTERS (<65%) ===
25ITR153 | Swetha | 24ITT32 | Attendance: 50.0% | Grade: SA
```

DATABASE:

STUDENTS DATABASE:

```
mysql> SELECT *FROM students;
+----------+--------------+------------+------------------+-------+-------+
| roll_no  | student_name | department | current_semester | email | phone |
+----------+--------------+------------+------------------+-------+-------+
| 25CSR120 | Deepeshwar   | CSE        |                3 | NULL  | NULL  |
| 25CSR173 | Mithun       | CSE        |                3 | NULL  | NULL  |
| 25ECR034 | Janani       | ECE        |                3 | NULL  | NULL  |
| 25ECR036 | Jaidharshini | ECE        |                3 | NULL  | NULL  |
| 25ECR045 | Lisha        | ECE        |                3 | NULL  | NULL  |
| 25ITR121 | Sahana       | IT         |                3 | NULL  | NULL  |
| 25ITR143 | Subadharsini | IT         |                3 | NULL  | NULL  |
| 25ITR144 | Subasree     | IT         |                3 | NULL  | NULL  |
| 25ITR153 | Swetha       | IT         |                3 | NULL  | NULL  |
| 25ITR164 | Trishna      | IT         |                3 | NULL  | NULL  |
| 25ITR172 | Vikasini     | IT         |                3 | NULL  | NULL  |
| 25MER001 | Akila        | MECH       |                3 | NULL  | NULL  |
| 25MER163 | Thulasi      | MECH       |                3 | NULL  | NULL  |
+----------+--------------+------------+------------------+-------+-------+
13 rows in set (0.00 sec)
```

COURSES DATABASE:

```
mysql> SELECT *FROM courses;
+-------------+--------------------------------------+---------+-------------+------------+
| course_code | title                                | credits | course_type | department |
+-------------+--------------------------------------+---------+-------------+------------+
| 24ADC31     | Artificial Intelligence & Knowledge  |     4.0 | EMBEDDED    | IT         |
| 24CEC31     | Surveying and Geomatics              |     4.0 | EMBEDDED    | IT         |
| 24CEL61     | Environmental Engineering Laboratory |     1.0 | PRACTICAL   | IT         |
| 24CEP71     | Project Work Phase 1                 |     3.0 | PRACTICAL   | IT         |
| 24ITC33     | ITC                                  |     3.0 | THEORY      | IT         |
| 24ITT32     | Programming in Java                  |     3.0 | EMBEDDED    | IT         |
+-------------+--------------------------------------+---------+-------------+------------+
6 rows in set (0.00 sec)
```

ENROLLMENTS DATABASE:

```
mysql> SELECT *FROM enrollments;
+---------------+----------+-------------+----------+-------------+-----------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | classes_held | classes_attended | medical_condonation |
+---------------+----------+-------------+----------+-------------+-----------------+---------------------+
|             2 | 25ITR143 | 24ITT32     |        3 |          50 |              48 |                   1 |
|             3 | 25ITR153 | 24ITT32     |        3 |          50 |              25 |                   0 |
|             4 | 25ITR172 | 24ITT32     |        3 |          50 |              42 |                   0 |
|             5 | 25MER163 | 24CEP71     |        3 |          45 |              41 |                   0 |
|             6 | 25ITR121 | 24ITC33     |        3 |          40 |              38 |                   1 |
+---------------+----------+-------------+----------+-------------+-----------------+---------------------+
5 rows in set (0.00 sec)
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

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9. YouTube, Java JDBC with MySQL Tutorial Videos. https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

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GITHUB:

LINK:	https://github.com/21suba07dharsini08/java_jdbc